

Comprehensive Analysis of 2025–2026 IEEE Quantum Computing Literature: Foundational Papers and Strategic Extension Pathways for Computer Science Researchers

The Evolving Convergence of Computer Science and Quantum Engineering

The landscape of computational research has undergone a profound paradigm shift throughout 2025 and 2026, driven by the United Nations' designation of 2025 as the International Year of Quantum Science and Technology¹. Quantum computing has transitioned from theoretical physics into a highly applied engineering discipline, increasingly intersecting with classical computer science domains such as artificial intelligence, software engineering, and network architecture¹. For computer science researchers entering this field, the current technological epoch—defined by Noisy Intermediate-Scale Quantum (NISQ) devices—presents a specific set of constraints and opportunities. Because large-scale, fault-tolerant quantum computers capable of continuous error correction remain on the horizon, current quantum algorithms cannot neglect the noisy nature of the hardware². Consequently, the most viable and rapidly expanding research pathways involve hybrid classical-quantum systems³. In these paradigms, classical computing infrastructure manages large-scale data processing, heuristic search, and overall system orchestration, while quantum processors are delegated to execute specialized subroutines such as high-dimensional feature mapping, non-linear transformations, combinatorial optimization, or quantum kernel evaluations³.

For a computer science researcher aiming to publish in high-impact IEEE venues, developing extensions to recently published, foundational papers offers a highly structured and effective entry point. A core theme emerging across 2026 literature is the convergence of generative artificial intelligence (GenAI), distributed quantum systems, and quantum software engineering⁴. By leveraging established classical computer science paradigms—such as automated software testing methodologies, cloud-native serverless architectures, evolutionary optimization algorithms, and deep learning attention mechanisms—researchers can create high-value, publishable extensions to existing quantum literature without requiring deep expertise in quantum mechanics or hardware fabrication. This report provides an exhaustive analysis of recent IEEE publications (2025–2026) that sit at the intersection of computer science and quantum engineering. It dissects prime publication venues, categorizes

foundational papers across Quantum Machine Learning (QML), Quantum Software Engineering (QSE), and Quantum Networking, and outlines rigorous, actionable pathways for extending these works into novel research.

Premier IEEE Publication Venues for Quantum Computer Science

A strategic publication plan requires precise alignment with the appropriate conferences and journals. The IEEE ecosystem offers highly visible platforms specifically tailored to quantum computing and engineering, with clear tracks designed to accommodate classical computer scientists.

IEEE International Conference on Quantum Computing and Engineering (QCE)

Commonly known as IEEE Quantum Week, the IEEE International Conference on Quantum Computing and Engineering (QCE) has rapidly grown into a premier international forum. Participation increased by more than 13% in 2025, scaling from 1,600 attendees in Montréal to over 1,750 in Albuquerque⁶. The 2026 iteration (QCE26) will be held at the Metro Toronto Convention Centre from September 13 to September 18, 2026⁶. QCE26 is particularly accommodating for computer science researchers, as it features a multidisciplinary structure encompassing multiple technical paper tracks directly aligned with classical computing subfields⁹.

The technical program tracks at QCE26 include Quantum Algorithms (QALG), Quantum Applications (QAPP), Quantum Machine Learning (QML), Quantum System Software (QSYS), and Quantum Networking & Communications (QNET)⁹. Furthermore, the introduction of specialized tracks such as Quantum-GenAI Co-Design & Co-Discovery (QGDD) reflects a high demand for research integrating large language models and generative architectures with quantum workflows⁵. The submission deadline for full technical papers is traditionally set for late April (e.g., April 27, 2026, for QCE26)⁹. The peer review process is highly rigorous; QCE25 received over 600 technical paper manuscripts, and organizers expect up to 700 for QCE26⁶. The review process utilizes EasyChair's bidding feature, requiring at least three Technical Program Committee (TPC) members to review each manuscript, ensuring that submissions are evaluated by experts with closely aligned domain knowledge⁶.

IEEE Transactions on Quantum Engineering (TQE)

For researchers targeting journal publications, the IEEE Transactions on Quantum Engineering (TQE) serves as the primary open-access venue within the IEEE ecosystem¹¹. TQE publishes regular, review, and tutorial articles based on the engineering applications of quantum phenomena¹¹. The journal specifically emphasizes quantum software, quantum communications, and hybrid classical-quantum computing models¹¹. Notably, the best technical papers from the QCE conference are routinely recommended for expanded

publication in TQE¹⁰, making it an ideal long-term target for extended conference papers. The journal operates under a single-anonymous peer review process and utilizes open-access Article Processing Charges (APCs) supported by various IEEE societies, including the Computer Society and the Signal Processing Society¹².

Publication Venue	Format	Next Key Cycle	Core CS Alignment Tracks	Evaluation Characteristics
IEEE QCE26 (Quantum Week)	Conference	Sept 13-18, 2026 (Toronto)	QML, QSYS, QNET, QGDD, QAPP	Minimum three expert reviews via TPC bidding; high acceptance standards for hybrid algorithms.
IEEE TQE	Journal	Rolling publication (2026)	Quantum Software, Machine Learning, Networking	100% Open Access; single-anonymous review; allows expanded versions of conference papers.
IEEE QCNC 2026	Conference	Apr 6-8, 2026	Quantum Networking and Computing	Double-blind review; specific focus on routing, distributed states, and network orchestration.

Advancements in Quantum Machine Learning (QML)

Quantum Machine Learning represents the most accessible entry point for computer scientists transitioning to quantum research, as it heavily relies on established classical neural network

architectures and dataset processing pipelines. Recent papers demonstrate that parameterized quantum circuits (PQCs) can replace or augment specific classical layers to enhance expressivity, reduce parameter counts, and improve robustness against noise and adversarial inputs.

Quantum Squeeze-and-Excitation (QSE) Networks

Classical Squeeze-and-Excitation (SE) modules have significantly improved neural network performance by adaptively recalibrating channel-wise feature responses, effectively simulating human visual attention mechanisms¹³. However, these classical modules often introduce substantial parameter overhead in deep networks. The foundational paper *Quantum Squeeze-and-Excitation Networks: Harnessing Quantum Noise for Robust Attention Mechanisms in Hybrid Architectures* by Peng et al. (IEEE TQE, June 2026) introduced the Quantum Squeeze-and-Excitation (QSE) Network, which replaces the fully connected layers of classical SE modules with parameterized quantum circuits¹³. This hybrid quantum-classical design utilizes quantum amplitude coding for data encoding, which drastically reduces the parameter count¹³.

A critical theoretical breakthrough in this paper is the treatment of quantum noise. Rather than viewing noise as purely destructive, the researchers demonstrated that quantum noise—specifically amplitude damping, depolarizing, phase-flip, and bit-flip noise—acts as a stochastic regularizer that mitigates the "Local Optima Trapping" problem caused by circuit redundancy¹³. To optimize the quantum circuits within the QSE module, the authors explored five different Controlled-NOT (CNOT) gate topologies: circular, linear, star, tree, and mesh¹³. Experimental evaluations on the CIFAR-10 dataset using a linear topology-based QSE ResNet-18 achieved 85.00% accuracy, outperforming the classical SE ResNet-18 baseline of 82.20% even in noiseless conditions, and maintaining above 85.00% performance across the four simulated noise models¹³. Subsequent applications of this module have successfully been deployed in time-conditioned U-Net architectures for satellite image cloud removal. Utilizing the RICE1 dataset, which provides paired cloudy and cloud-free optical images, the QSE-enhanced U-Net consistently improved reconstruction quality in terms of peak signal-to-noise ratio (PSNR) and structural similarity index measure (SSIM)¹⁴.

For a computer science researcher, extending QSE networks provides numerous accessible pathways. The original QSE paper applies quantum attention primarily to Convolutional Neural Networks such as ResNet-18 and U-Net¹³. A highly promising extension involves adapting the QSE module for Transformer-based architectures, such as Vision Transformers (ViTs) or lightweight Large Language Models (LLMs). A researcher could systematically replace the classical multi-head self-attention bottleneck layers with QSE modules, establishing a comparative analysis to determine whether the parameter-efficiency and noise-induced regularization observed in CNNs hold true for sequence-to-sequence tasks. Another significant extension involves adversarial robustness evaluation. Since the foundational paper establishes that quantum noise provides regularization¹³, a natural extension is to evaluate QSE networks against adversarial attacks, such as the Fast Gradient Sign Method (FGSM) or Projected Gradient Descent (PGD). The research hypothesis would propose that the inherent

stochasticity of the quantum noise models utilized in QSE provides superior defense mechanisms against adversarial perturbations compared to classical dropout techniques.

Quantum Cross-Hybrid Fine-Tuning (QCHFT) for Large Language Models

As foundational language models scale exponentially, full-parameter fine-tuning becomes computationally prohibitive, necessitating Parameter-Efficient Fine-Tuning (PEFT) methods like Low-Rank Adaptation (LoRA)¹⁶. However, classical PEFT methods are constrained to linear transformations and often struggle to capture complex, high-order feature interactions under stringent parameter budgets¹⁶. Addressing this limitation, Li et al. published *QCHFT: Quantum Cross-Hybrid Fine-Tuning for LLMs* (IEEE TQE, 2026), proposing a suite of PQC-augmented adapters: the Matrix Product State (MPS) PQC, Alternating and Hierarchical Pairwise Contraction (APC/HPC) PQC, and LinearA–Quantum–LinearB (AQB) PQC¹⁶.

By dividing feature representations into multiple small quantum circuits surrounded by learnable linear projections, QCHFT increases per-parameter expressivity via quantum entanglement while maintaining a minimal parameter footprint¹⁶. The QCHFT framework relies

on an automatic qubit scaling logic where the number of qubits (n) is dynamically determined by the maximum dimensionality between input and output features, formulated as

$$n = \lceil \log_2(\max(d_{in}, d_{out})) \rceil$$
¹⁹. This ensures features are perfectly mapped into the quantum Hilbert space without capacity bottlenecks¹⁹. Furthermore, the model utilizes

rank-controlled quantum measurements, where a rank parameter (r) dictates the dimension

of the quantum features, requiring measurements on the last r qubits of the circuit (where

$r \leq n$)¹⁹. Tested across multiple causal LLM backbones—including Mamba2-130M, GPT-Neo-125M, Pythia-160M, Llama-3.2-1B, and Qwen-3.5-2B—on datasets such as a multi-tier IoT resource-allocation task, SST-2, and HellaSwag, QCHFT consistently outperformed standard LoRA¹⁶. For instance, on Mamba2-130M, the APC–PQC variant improved accuracy from 67.33% with classical LoRA to 72.66%¹⁶.

Extending QCHFT presents an excellent opportunity for researchers familiar with natural language processing. One critical area of exploration is addressing the barren plateau phenomenon within the QCHFT adapters. Deep parameterized quantum circuits suffer from barren plateaus, where gradients vanish exponentially, hindering the training process²¹. While QCHFT utilizes shallow circuits to mitigate this, an extension could integrate advanced classical initialization strategies—such as Reinforcement Learning (RL)-based initialization or Gaussian sampling distributions—directly into the QCHFT training pipeline²². The research could evaluate whether RL-initialized QCHFT adapters allow for deeper quantum circuits without succumbing to vanishing gradients, thereby unlocking even higher expressivity for complex linguistic tasks. A second extension pathway involves cross-modal quantum adapters. Researchers could extend the QCHFT framework beyond text-based LLMs to multimodal models, such as

Vision-Language Models. By utilizing the tensor-network topologies explored by Li et al., researchers can test whether quantum adapters are exceptionally suited for aligning discordant embedding spaces (e.g., aligning image tokens with text tokens) in low-resource environments.

Integrated Encoding and Quantization for Quanvolutional Neural Networks

Quanvolutional Neural Networks (QuanvNNs) utilize quantum circuits as random feature extractors for image processing, offering a highly NISQ-friendly approach by avoiding the need to optimize complex quantum circuit parameters via the computationally expensive parameter shift rule²⁴. Traditional preprocessing pipelines for quanvolutional layers involve four rigid steps: binary quantization, state encoding, circuit processing, and decoding²⁴. The foundational paper *Integrated Encoding and Quantization to Enhance Quanvolutional Neural Networks* by Bosco et al. (IEEE TQE, December 2025/2026) identified that binary quantization causes severe information loss for tasks requiring fine-grained detail, and that disjointed encoding and processing steps consume unnecessary quantum resources²⁴.

To resolve these inefficiencies, the authors introduced a flexible data quantization approach

coupled with a memoization lookup table. The quantization formula, $q(x) = \frac{\lfloor x \cdot N \rfloor}{N-1}$, allows

the analytical selection of quantization levels (N) based on the task's specific information retention requirements²⁴. This enables a deliberate tradeoff between retaining detail (higher

N) and lowering computational load²⁶. Furthermore, the authors introduced an *integrated encoding strategy* that fuses the encoding and processing steps into a single quantum circuit, which significantly reduces circuit depth²⁴. Evaluated on benchmark datasets such as MiraBest and LArTPC, the integrated approach proved substantially more resource-efficient than traditional rotational encoding, allowed for the testing of larger kernel sizes, and outperformed classical Convolutional Neural Networks (CNNs) across multiple classification tasks²⁴.

The most logical extension of this work for a classical computer scientist involves dynamic memoization in streaming data environments. The memoization technique proposed by Bosco et al. fills a lookup table dynamically during dataset preprocessing²⁶. An extension could apply this architecture to real-time video stream processing or continuous reinforcement learning environments. The research would formulate a classical algorithm to dynamically prune, update, and manage the quantum memoization table to strictly maintain memory bounds while processing temporal image data. Additionally, researchers could explore hardware-aware integrated encoding. The flexible nature of the integrated encoding scheme allows it to be adapted to specific quantum hardware topologies, such as heavy-hexagon superconducting grids or fully connected trapped-ion systems. A paper could optimize the integrated encoding circuits specifically for restricted topologies, minimizing the need for SWAP gates during transpilation, and benchmarking the resulting performance improvements using simulators like Qiskit Aer GPU or physical backends⁵.

Foundational QML Architecture	Identified Classical Problem	Quantum-Enhanced Solution	Promising Extension Vectors
QSE Networks [cite: 13]	High parameter cost of SE attention layers; vulnerability to perturbations.	PQC substitution; leveraging quantum noise as a stochastic regularizer.	Integration into Vision Transformers; adversarial defense testing against FGSM/PGD attacks.
QCHFT Adapters [cite: 16]	Suboptimal performance and low expressivity of classical linear PEFT (LoRA).	Tensor-network based PQC adapters (MPS, APC) with automatic qubit scaling.	Multimodal adapter alignment; RL-driven barren plateau mitigation for deeper circuits.
QuanyNN (Integrated) [cite: 24]	Information loss in binary quantization; high circuit depth due to disjointed steps.	Variable N -level quantization with memoization; fused encoding and processing circuits.	Hardware-aware topology mapping; dynamic temporal memoization for video streaming.

Quantum Software Engineering, Circuit Optimization, and Error Mitigation

While Quantum Machine Learning focuses on algorithmic applications, Quantum Software Engineering (QSE) addresses the compilation, execution, testing, and optimization of quantum programs. This thematic area is a highly fertile ground for traditional software engineers, as the underlying challenges closely mirror classical compiler optimization, distributed systems architecture, and automated software testing.

Variance Reduction in Quantum Circuit Cutting

Noisy intermediate-scale quantum (NISQ) devices suffer from severely limited qubit counts and restricted hardware connectivity, limiting the scale of executable quantum circuits. Circuit cutting, also known as circuit knitting, is a compilation technique that simulates larger quantum computers by decomposing a massive circuit into smaller subcircuits that fit onto available hardware²⁸. By applying quasiprobabilistic sampling and classical post-processing, the expectation value of the original large circuit can be accurately reconstructed, albeit at the cost

of a sampling overhead that scales exponentially with the number of cuts³¹. Existing circuit cutting methods focus primarily on fitting the subcircuits within the qubit count limits of the target hardware but neglect the resulting depth of the subcircuits after hardware routing²⁸. Because quantum state fidelity decays exponentially with circuit depth, overly deep routed subcircuits render the final execution useless²⁸.

To address this, Eslaminia and Le Beux published *Reducing Maximum Subcircuits Depth in Quantum Circuit Cutting* (IEEE TQE, January 2026), formulating the initial circuit cutting problem as a multi-objective optimization problem based on an evolutionary search method²⁸. The fitness objectives included minimizing both the total number of cuts (to strictly control sampling overhead) and the maximum depth of the individual routed subcircuits²⁸.

Implemented as an extension to the existing Qiskit Circuit Cutting Addon, their approach achieved reductions in routed subcircuit depth of up to 9.9% across standard benchmark circuits compared to baseline methods²⁸.

Extending circuit cutting optimization provides an excellent avenue for applying classical artificial intelligence to quantum compilers. Evolutionary algorithms, while effective, can be computationally expensive and slow to converge for massive circuits. An extension could replace the evolutionary search mechanism with a Graph Neural Network (GNN) trained on circuit Directed Acyclic Graphs (DAGs) to predict optimal cut locations instantly. This would bridge classical AI with quantum compiler optimization, offering a low-latency alternative to evolutionary multi-objective optimization. Furthermore, once a circuit is cut, the resulting subcircuits and the intensive classical post-processing tasks must be orchestrated. A researcher could design a distributed serverless orchestration framework specifically tailored for quantum circuit cutting workflows³⁰. This research would involve modeling the execution time, queue latency, and classical reconstruction overhead to mathematically minimize the end-to-end makespan across a hybrid quantum-HPC cluster.

Advanced Error Mitigation via Pauli Check Extrapolation

Error mitigation is an absolute necessity in the NISQ era to extract meaningful signals from noisy hardware. A popular technique is Zero-Noise Extrapolation (ZNE), which artificially amplifies circuit noise through the insertion of logical identity operations and then mathematically extrapolates the measured expectation values back to a theoretical zero-noise limit³³. However, ZNE requires artificially increasing noise levels, an approach that fails catastrophically for circuits that are already highly noisy³³. Alternatively, Pauli Check Sandwiching (PCS) is an error detection scheme that inserts pairs of parity checks and uses post-selection to discard execution runs that signal errors via an ancilla qubit³³. Unfortunately, physically inserting additional check layers introduces more physical noise and exponentially decreases the post-selection success rate³³.

In the paper *Extrapolating Pauli Checks for Expectation Value Estimation on Noisy Quantum Devices* (IEEE TQE, May 2026), Langfitt et al. introduced Pauli Check Extrapolation (PCE). Instead of extrapolating to zero noise by adding noise, PCE obtains measured expectation values from circuits with varying (but small) numbers of physical parity checks, and extrapolates to the theoretical "maximum check" limit—the mathematical point where unit

fidelity is achieved³³. Utilizing both linear ($E(n) = \alpha + \beta n$) and exponential ($E(n) = ab^n + c$) extrapolation models derived from Markovian error models, PCE substantially outperformed ZNE on large random Clifford circuits³³. On actual IBM hardware (the IBM Kingston backend), PCE achieved an accuracy of up to 99.2%, compared to ZNE's 82%, for 4-qubit circuits³³. Notably, PCE eliminates the need for the extensive calibration procedures required by state-of-the-art robust shadow (RS) estimation protocols, drastically reducing the overall sampling size³³.

A primary limitation of PCE is its strict restriction to Clifford or near-Clifford circuits where valid Pauli checks exist³³. A highly impactful extension would involve developing a compiler pass that automatically identifies maximal Clifford subcircuits within non-Clifford algorithms, such as the Variational Quantum Eigensolver (VQE) or Quantum Approximate Optimization Algorithm (QAOA)³³. The framework would apply PCE locally to the Clifford blocks while utilizing probabilistic error cancellation (PEC) or ZNE for the non-Clifford gates, creating a composite, heterogeneous error mitigation pipeline. Additionally, researchers could integrate PCE with classical software testing heuristics. The extrapolated limits heavily depend on the initial expectation values. A researcher could integrate PCE with dynamic testing to automatically verify the bounds of the extrapolation model, dynamically adjusting the extrapolation curve from linear to exponential if anomalous outputs are detected.

Dynamic Testing Strategies for Quantum Programs

Traditional software testing methodologies, while highly effective for classical programs, fail dramatically when applied to quantum programs due to the inherent characteristics of quantum systems, such as superposition, entanglement, and the exponentially expanding input space³⁶. Most contemporary quantum testing studies limit test inputs strictly to pure states, ignoring mixed states (which represent probabilistic mixtures of pure states), leading to incomplete coverage and exorbitant time costs for achieving input domain coverage³⁶. Furthermore, traditional strategies do not account for the hidden, non-deterministic failure patterns associated with qubit operations³⁶.

Huang et al. addressed these deficiencies in *A Dynamic Testing Strategy With Incremental Learning Model for Quantum Programs* (IEEE TQE, October 2025/2026)³⁶. The researchers introduced Quantum Dynamic-Testing with Incremental Learning (QDT-IL), a novel approach that expands the input domain to include mixed states for the better utilization of quantum information³⁶. QDT-IL employs an incremental learning model to continuously learn from executed test cases and update its predictions regarding failure tendencies for new, untested inputs³⁶. By utilizing distance-based diversity metrics, the strategy systematically increases test case variety, driving the targeted exploration of failure-prone regions in the quantum input space³⁶. This adaptive approach demonstrated superior fault detection efficiency over baseline random testing strategies³⁶.

Extending this work provides profound opportunities for software engineering researchers. The QDT-IL model can be adapted specifically for testing Variational Quantum Algorithms (VQAs).

Unlike static quantum programs, VQAs evolve continuously during training. An extension would involve modifying QDT-IL to test the gradient robustness of a VQA at various epochs, using the incremental learning model to rapidly detect testing inputs that trigger barren plateaus or local minima traps during the optimization loop. Another significant extension is the introduction of a framework for quantum mutation testing. By injecting subtle unitary transformations (representing developer bugs) into the quantum code, a researcher could evaluate how rapidly the QDT-IL algorithm can generate mixed-state test cases capable of "killing" these specific quantum mutants compared to traditional pure-state fuzzing techniques³⁶.

Quantum Systems, Orchestration, and Networking

Applying quantum algorithms to classical telecommunications, cloud orchestration, and networking infrastructure represents a vital bridge between theoretical quantum optimization and immediate industry utility.

Hybrid Classical-Quantum Serverless Computing Architectures

As hybrid algorithms like VQAs mature toward practical utility in fields like chemistry and operations research, the tight integration of classical High-Performance Computing (HPC) and Quantum Processing Units (QPUs) requires highly sophisticated software orchestration³². Serverless computing, or Function as a Service (FaaS), is increasingly considered a promising solution due to its inherent scalability, lightweight containerized virtualization, and abstraction of resource management³². In a hybrid serverless architecture, classical HPC resources and QCs interact via stateless functions, dynamically scaling up workers as application demand increases³².

Cicconetti defined a rigorous system model for a hybrid classical-quantum serverless platform in *Modeling and Performance Evaluation of Hybrid Classical-Quantum Serverless Computing Platforms* (IEEE TQE, 2025)³². The study provided an open-source numerical simulator capable of being driven by production traces and stochastic workload models generated via IBM Qiskit³². The author evaluated various scheduling policies—specifically FIFO, Priority, and Mixed policies—for VQAs³². The simulation results revealed that a Mixed scheduling policy effectively balances performance, job completion rates, and resource utilization across classical containers and quantum task queues for jobs with highly heterogeneous problem sizes and priorities³².

Extending Cicconetti's orchestration model allows researchers to tackle advanced deployment constraints. A prime extension is evaluating the serverless overhead of extensive quantum error mitigation techniques. Because advanced error mitigation (like PCE or ZNE) requires launching exponentially many variant circuits and executing intensive classical post-processing, a scheduling algorithm optimized specifically for *latency-sensitive error mitigation workloads* in a multi-tenant cloud environment would be a highly novel contribution. Furthermore, while the foundational paper optimizes primarily for job completion and utilization⁴¹, modern HPC trends increasingly prioritize sustainability and energy efficiency. Research could introduce an energy-aware scheduler to the open-source simulator, mathematically trading off the energy

cost of running classical parameter updates against the energy cost of QPU execution, utilizing advanced metrics such as Energy-to-Solution (ETS)¹⁸.

Quantum-Based Resilient Routing for Telecommunications

Layer 3 (L3) network routing optimization operating over predefined Layer 1 (L1) optical links is a massively computationally intensive combinatorial problem⁴². Specifically, finding paths that minimize end-to-end latency while simultaneously avoiding correlated dual-link failures—often caused by shared physical infrastructure known as Shared Risk Link Groups (SRLGs)—requires balancing competing multi-objective quadratic terms⁴². When shared infrastructure fails, all traversing links fail simultaneously, rendering topologically disjoint paths inadequate for true resilience⁴². The inclusion of joint failure probabilities across edge pairs renders the problem computationally intractable for classical solvers as network size scales⁴².

Harb et al. addressed this in *Quantum-Based Resilient Routing in Networks: Minimizing Latency Under Dual-Link Failures* (IEEE TQE, 2026)⁴². The authors formulated this as a graph-based optimization problem aimed at finding two vertex-disjoint paths from edge secondary sites to the internet backbone (terminal sites), coupled with a novel resiliency penalty metric based on pairwise joint-failure probabilities that systematically avoids high-risk link combinations⁴². The mathematical formulation was adapted and solved using the Quantum Approximate Optimization Algorithm (QAOA) on both simulators and quantum hardware⁴². Tested on a toy graph topology with 5 vertices and 7 edges, QAOA successfully found optimal minimum-energy states for networks under both independent (uncorrelated) and highly dependent (correlated) failure scenarios, significantly outperforming classical approximations in identifying the safest routes⁴².

A major limitation of QAOA for network routing is the massive qubit overhead required to map dense, large-scale telecommunication graphs⁴³. By combining Harb et al.'s optimization formulation⁴⁴ with Eslaminia’s circuit cutting techniques²⁸, a researcher could partition large telecommunication graphs into smaller topological sub-graphs, execute QAOA on the sub-graphs independently on near-term hardware, and stitch the routing results together via classical heuristics. This hybrid approach would drastically scale the size of the networks that can be optimized on current NISQ devices. Additionally, while QAOA efficiently addresses static link failure probabilities, real-world network traffic is highly dynamic. An extension could adapt the Quadratic Unconstrained Binary Optimization (QUBO) formulation to account for real-time traffic flow dynamics, leveraging hybrid quantum annealing algorithms⁴³. This research would empirically evaluate how rapidly a hybrid solver can recalculate resilient paths when traffic loads shift abruptly, measuring Time-to-Convergence against classical Dijkstra or Suurballe algorithmic variations⁴².

Systems & Networking Challenge	Foundational Approach	Proposed CS Extension Vector
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Serverless Resource Orchestration	Hybrid scheduling (FIFO/Priority/Mixed) for VQAs ³² .	Energy-aware (ETS) scheduling for massive, latency-sensitive error mitigation workloads.
L3 Latency-Resilient Routing	QAOA with dual-link failure penalty metrics ⁴² .	Scaling QAOA routing via circuit cutting; dynamic traffic routing using hybrid quantum annealing.
Quantum Program Verification	Incremental Learning with mixed states via QDT-IL ³⁶ .	VQA gradient robustness testing and automated quantum mutation testing frameworks.

Strategic Methodological Guide for Computer Science Researchers

For a computer scientist entering the quantum domain with the intention of publishing in high-tier venues like IEEE QCE26 or IEEE TQE, adopting a structured, pragmatic research methodology is imperative. The most successful approach for a beginner in this field is not to attempt the invention of fundamentally new quantum physics or novel hardware architectures, but rather to systematically apply rigorous classical computer science paradigms to existing quantum bottlenecks.

1. **The Modular Replacement Strategy:** Identify a classical machine learning or software engineering pipeline and replace a highly specific, computationally complex module with a parameterized quantum circuit. The QSE Networks¹³ and QCHFT LLM adapters¹⁶ serve as perfect templates for this methodology. The core of the research extension lies in empirically proving that this specific replacement offers a distinct, measurable advantage—be it exponential parameter reduction, inherent noise resilience, or superior feature expressivity.
2. **The Cross-Pollination of Solutions:** Profound innovation frequently occurs at the intersection of two distinct quantum subfields. For example, applying techniques from Quantum Software Engineering, such as circuit cutting²⁸, to constraints in Quantum Machine Learning, such as reducing the depth of a QuanyvNN integrated encoding circuit²⁴, generates an immediate, highly publishable research narrative. Similarly, applying classical software testing heuristics, such as the QDT-IL model³⁶, to evaluate the bounds of quantum error mitigation protocols like PCE³³ elegantly bridges software testing and hardware optimization.
3. **Prioritization of Simulation and Benchmarking:** Because physical access to

large-scale, noise-free QPUs remains severely limited, IEEE venues highly value robust theoretical models paired with extensive, reproducible simulation. Utilizing open-source environments like Qiskit Aer to simulate realistic noise models (e.g., amplitude damping, phase-flip) and proving that a proposed software architecture—such as a hybrid serverless scheduler³² or a new QCHFT adapter¹⁶—improves end-to-end metrics is a scientifically valid and highly respected approach¹³.

Conclusion

The exhaustive literature published across 2025 and 2026 clearly indicates a fundamental paradigm shift: quantum computing is rapidly evolving into applied quantum computer science. The rigid barriers that once separated theoretical quantum physics from classical software engineering are rapidly dissolving, replaced by hybrid architectures, co-design philosophies, and hardware-aware software optimizations².

By strategically targeting the direct intersections of classical and quantum computing—such as Quantum Squeeze-and-Excitation attention mechanisms for image restoration¹³, Quantum Cross-Hybrid Fine-Tuning adapters for foundational Large Language Models¹⁶, advanced multi-objective circuit cutting for NISQ compilation²⁸, and dynamic software testing frameworks utilizing mixed states³⁶—a beginning researcher can build directly atop the vanguard of current literature. Premier venues like IEEE QCE26 and IEEE TQE actively seek contributions that translate these theoretical quantum advantages into practical, engineered realities⁷. By rigorously executing the extension pathways detailed in this report, computer science researchers can confidently contribute high-impact solutions that advance the integration of quantum systems into the global computational infrastructure, driving the industry toward the realization of practical quantum utility.

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